

STAT 425 Project Proposal

March 6th, 2026

Zoya Malik, 30139545

Manreet Kaur Jhattu, 30120808

Lachlan Findlay, 30249420

Experimental Design:

Our experiment is to individually undergo a typing test under auditory factors such as lo-fi music, upbeat music, white noise, and silence. We aim to investigate typing speed and accuracy under different treatment levels in order to determine whether there is an effect that is statistically significant. We will be conducting three typing tests of one auditory condition per day. These will take place at the same time each day until we have done two full rounds of all four auditory conditions. We will be doing two full rounds to account for day to day extenuating factors. This yields $(3 \text{ participants}) \times (4 \text{ treatments}) \times (3 \text{ trials per treatment session}) \times (2 \text{ replications}) = 72$ individual typing trials. Therefore, each treatment will have 18 trials.

We will be carrying out a two-factor experiment as we have one auditory factor and one blocking factor (subject). Each new day, we will use a randomized generator to choose a new treatment level for each member of our group to conduct their three consecutive typing tests. This will continue until all four treatments have been chosen. We will carry out a Complete Randomized Block Design. Each person (the block) completes all treatments and the order of conditions is independently randomized in sequence. We will not be using a Latin Square design as this is appropriate if we had a second blocking variable. Since there is only one blocking variable, the subject, the Complete Randomized Block Design is sufficient.

The auditory condition is a fixed factor. The four levels (lo-fi music, upbeat music, white noise, and silence) were deliberately selected and represent the only conditions we wish to compare, not a random sample from a larger set of possible audio environments. The blocking factor (subject) is also fixed, since the three participants are the specific individuals conducting the experiment rather than a random sample drawn from a broader population.

Lastly, the trials will be randomized as we will be shuffling a respective playlist for each time we do the test. This is so that we are accounting for any bias from individual songs. All participants will use a standardized volume level of approximately 60–65 dB (a comfortable listening volume) across all music and white noise conditions, to ensure that any observed differences are due to the type of audio content rather than the volume. The same pair of headphones will be used by all participants throughout the experiment to eliminate headphone/earbud variation as a confounding factor. During the silence condition, participants will still wear the headphones with no audio playing, so that the physical sensation of wearing headphones is controlled across all four conditions. To further ensure homogeneity across trials, each participant will use the same keyboard and computer setup for all of their trials. Trials will be conducted in a quiet, distraction-free environment.

Experimental units/subjects:

We will be participating in this study therefore the subjects will be all three of our group members. Each of us will be typing with each treatment factor and undergoing every trial individually. Each experimental unit is a single one-minute typing trial. The data point collected from each trial is the WPM score recorded by [typing.com](https://www.typing.com) at the end of that one-minute test.

Response Variable:

Our response variable is words per minute (WPM). WPM is a continuous ratio-scale measurement making it suitable for analysis with ANOVA without any transformation or categorization necessary. The raw WPM is recorded for each trial as converting the data into categories would eliminate the differences we are trying to detect.

Software and Storage:

The software program we will be using is Rstudio. The data will be stored in a csv file so it can be imported and analyzed in RStudio. The raw data will be stored securely in a shared Google Drive folder accessible only to the three group members. No external parties will have access to the data. A backup copy will be maintained on each member's local machine. We will be using the website [typing.com](https://www.typing.com) to test the typing speed.